

Colouring Graphs with forbidden K_7 -vertex minors Feb 24 2026

4CT: A&H 1976, every planar graph is 4-colourable

- planar $\Rightarrow$ no K_5 , $K_{3,3}$; Kuratowski/Wagner

no K_5 minor: decomposes into planar Mobius ladder

Hindman's conjecture: every graph w/ no K_t minor is $(t-1)$ -colourable.

$t=2$: $\Rightarrow$ 1 colourable, edgeless

$t=3$: $\Rightarrow$ no cycle, bipartite so 2 colourable

$t=4$: $\Rightarrow$ series-parallel graphs (degenerate) $\Rightarrow$ 3 colourable

$t=5$: $\Rightarrow$ equivalent to 4CT

$t=6$: $\Rightarrow$ reduces to 4CT (Robertson, Seymour, Thomas)

$t=7$: open

Relaxation: colours
every class w/ no K_{7-} is 8 colourable (Usser, Conzelmann).

Relaxation: minors:

$$K_7^- : \begin{array}{|c|} \hline \cdot \\ \hline \end{array} : K_7^v : \begin{array}{|c|} \hline \cdot \\ \hline \cdot \\ \hline \end{array} : K_7^E : \begin{array}{|c|} \hline \cdot \\ \hline \cdot \\ \hline \cdot \\ \hline \end{array}$$

[Jakobsen] every graph with no K_7^v and no K_7^E is 6-colourable.

26.

Theorem: [Norin, T]

Every graph w/ no K_7 -minor is 6-colourable

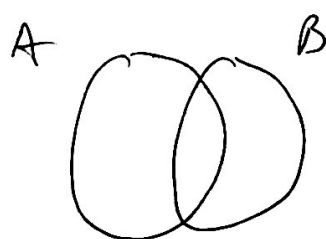
lemma: every 4-^{vertex}-connected graph G with $|E(G)| \geq |V(G)| - 8$

then G has no K_7 -minor unless G is $K_{2,2,2,2}$

$$n=8: n-8=0 \quad \binom{8}{2}=28$$

claim: a minor-minimal counter-example G is 5-connected

proof: by contradiction

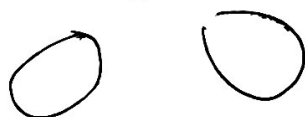


let $k = |E(G[A \cap B])|$, if K_4 's
contracted from B onto $A \cap B$

$$|E(A)| + (6-k) \leq 4|A| - 9$$

$$\begin{aligned} |E(G)| &\leq 4(|V(G)| + 4) - 2(6-k) - 2 + 9 - k \\ &\leq 4|V(G)| - 9 \quad \text{for } k \leq 5 \end{aligned}$$

Case: $k=6$

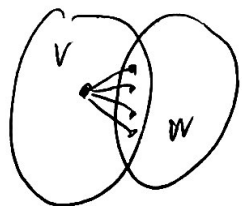


Case: cannot contract K_4 from B [Robertson, Seymour, Thomas]

still get K_7 and bound

Case: every 4 separation has a single vertex on one side

all of $M(v)$ has deg 4



$$\Rightarrow |E(G)| \leq 2|V(G)|$$

contract one edge, get G'

$$\begin{aligned} |E(G')| &\leq 4|V(G)| - 9 \\ &\leq 4|V(G)| + 9 - 9 \end{aligned}$$

$\Rightarrow$ every edge is in at least 4 triangles.

delete one edge, get G'

$$|E(G')| = |E(G)| - 1 \leq 4|V(G)| - 8$$

$$\text{so } |E(G)| = 4|V(G)| - 7$$

$$\text{so } \deg(G) \leq 7$$

$$\deg(G) \leq 5$$

$\deg(G) = 6$: 2-linkages

$\deg(G) = 7$: get K_7^v minor

[Kurabayashi, Tait]: every graph w/ no K_7 or $K_{4,4}$ minor is 6-colourable

proof: a minor minimal counterexample is a graph G

- has no K_7^V -minor

- $\chi(G) \geq 7$, but any proper minor G' has $\chi(G') \leq 6$.

k -contraction critical graphs has $\chi(G) \geq k$, but any proper minor G' has $\chi(G') \leq k-1$

$\Rightarrow 7$ connected [Dirac]

$\Rightarrow \delta(G) \geq 7$

$\Rightarrow |E(G)| \leq 4|V(G)| - 9$

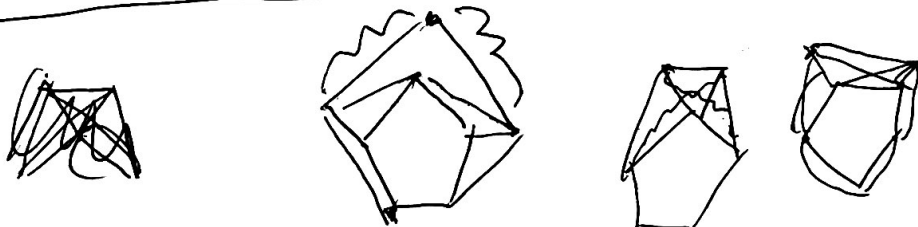
$\Rightarrow 18$ vertices of deg. 7

~~Case 1~~ N is

- independence number α

$\alpha(G[N(V)]) \leq 2 \leq \deg(v) - k + 2$

Case: N is connected must be an inflation of a 5 cycle



N is not connected $= K_3 \vee K_4 \Rightarrow 7, 4$ distinct K_5

open: every graph w/ no K_7^V minor is 6-colourable